Hydration and Nutrition for Physical Activity

Why Hydration Matters

Water makes up roughly 50 to 60 percent of total body weight and is involved in nearly every bodily function, including regulating temperature, transporting nutrients, lubricating joints, and removing waste products. Even mild dehydration, a loss of as little as 2 percent of body weight in fluid, can impair physical performance and cognitive function.

Daily Fluid Needs

General guidelines suggest adults consume roughly 2.7 to 3.7 liters of total water per day from all sources, including beverages and food, though individual needs vary based on body size, climate, and activity level. Thirst is a reasonably reliable indicator of hydration needs for most healthy people, although it can lag behind actual fluid deficits during intense exercise.

Urine color is a practical, low-cost way to gauge hydration status: pale yellow generally indicates adequate hydration, while dark yellow or amber suggests a need to increase fluid intake.

Hydration During Exercise

During exercise, fluid needs increase to compensate for sweat losses. As a general guideline, athletes are often advised to drink 400 to 600 milliliters of fluid in the two to three hours before exercise, and to continue drinking small amounts, roughly 150 to 250 milliliters every 15 to 20 minutes, during prolonged activity lasting more than an hour.

For exercise sessions lasting longer than 60 to 90 minutes, especially in hot conditions, sports drinks containing electrolytes and carbohydrates can help maintain blood sodium levels and provide a quick energy source, reducing the risk of cramping and fatigue.

Pre- and Post-Workout Nutrition

Eating a meal containing both carbohydrates and a moderate amount of protein roughly two to three hours before exercise helps top off glycogen stores and provide steady energy. If eating closer to a workout, a smaller, easily digestible snack such as a banana or a slice of toast is generally better tolerated.

After exercise, particularly resistance training, consuming protein along with carbohydrates within a couple of hours supports muscle repair and glycogen replenishment. A common practical target is 20 to 40 grams of protein post-workout, though individual needs vary with training intensity and goals.

Electrolytes

Sodium, potassium, magnesium, and chloride are electrolytes lost through sweat during exercise. For most moderate exercise sessions under an hour, plain water is sufficient to replace fluid losses. For longer or more intense sessions, especially in hot or humid conditions, replacing electrolytes alongside fluids helps prevent cramping, dizziness, and a dangerous condition called hyponatremia, which results from excessive water intake without adequate sodium replacement.